Threshold Calibration Against a Human Reference Standard

Step-by-step calculation of every reported value

Phase 5 — Curriculum/Labour-Market Gap Analysis

29 July 2026

Contents

0. Notation and inputs 1

1. The evaluation set 2

2. Data collection 2

3. Cleaning 2

4. Pre-registered exclusions 3

5. Computing the MOS 3

6. Reliability of the MOS 3

7. Binarising the MOS 4

8. Classification and the confusion matrix 4

9. The metrics 4

10. Choosing the threshold — Youden's J 5

11. Area under the ROC curve 5

12. Uncertainty — bootstrap 5

13. Sensitivity — the strict claim 5

14. Result 6

15. Reproducibility notes 6

0. Notation and inputs

Symbol	Meaning	Source
s_i	cosine similarity of pair i	BGE-base-en-v1.5 encoder, frozen run `20260727T105055Z-475eb3`
x_{ij}	score given to pair i by rater j, integer $0-10$	blind human annotation
μ_i	**MOS** (mean opinion score) of pair i	computed, Step 5
y_i	binary gold label of pair i	computed, Step 7
t	candidate similarity threshold	swept, Step 11
$\hat{y}_i(t)$	predicted label at threshold t	$\mathbb{1}[s_i \geq t]$

Two inputs only: the **cosine scores** produced by the method under test, and the **human scores**. No language model contributes to any value in this document.

---

1. The evaluation set

$N = 250$ skill×course pairs, frozen and hash-locked (`sheet_hash = 78dfd3fe...f579a169`) before any rating began.

Block	n	Cosine range	Displayed by the system?	Sampling
Census	203	$0.662 - 1.000$	yes	**complete** — every cell above the floor
Boundary	17	$0.656 - 0.660$	yes (rounds up to 0.66)	complete
Controls	30	$0.434 - 0.655$	no	stratified sample

Consequence. Because the census is complete, *precision computed on displayed cells carries no sampling error*. Recall depends on the 30-pair control sample and is therefore an estimate.

---

2. Data collection

Each rater saw only the skill, the learning outcome and the course name — never the cosine, band or block. Scale $0-10$ with printed anchors at $0 =$ *not taught*, $5 =$ *the basics only*, $10 =$ *fully taught*. Pairs $1-30$ (the anchor block) were served first in identical order to every rater; the remaining 220 were shuffled per rater.

---

3. Cleaning

$$2001 \text{ rows} \; - \; 25 \text{ (test account)} \; - \; 1 \text{ (duplicate participant)} \; = \; 1975 \text{ rows}$$

The duplicate was detected by an `ext_id` collision across two different display names, i.e. one person submitting under two labels. Integrity checks passed: no repeated $(\text{rater}, \text{pair})$ entry, all scores integer in $[0, 10]$, all pair identifiers in $[1, 250]$.

---

4. Pre-registered exclusions

Rules fixed before data collection: exclude a rater if **(a)** median response time < 2 s, **(b)** anchor-block Spearman $\rho < 0.20$ against the other raters, or **(c)** fewer than 20 pairs rated.

One rater was excluded under (a) with a median of 1,962 ms.

$$8 \text{ raters} \;\longrightarrow\; K = 7 \text{ included} \quad (6 \text{ with complete data})$$

---

5. Computing the MOS

$$\mu_i \;=\; \frac{1}{K_i} \sum_{j=1}^{K_i} x_{ij}$$

Worked example — pair 15: *restful api development* ↔ *Hybrid App Development* (Introduction To Mobile Application Development), $s_{15} = 0.6547$.

$$x = [10, 6, 9, 6, 8, 6, 8] \qquad \mu_{15} = \frac{53}{7} = 7.571$$

---

6. Reliability of the MOS

Two-way random-effects ANOVA, absolute agreement, on the $n = 250$ pairs $\times$ $k = 6$ raters with complete data. Grand mean $\bar{x} = 5.583$.

$$MS_R = \frac{k \sum_i (\bar{x}_{i\cdot} - \bar{x})^2}{n-1} = 28.959 \qquad MS_C = \frac{n \sum_j (\bar{x}_{\cdot j} - \bar{x})^2}{k-1} = 462.382$$

$$MS_E = \frac{\sum_{ij} (x_{ij} - \bar{x}_{i\cdot} - \bar{x}_{\cdot j} + \bar{x})^2}{(n-1)(k-1)} = 4.395$$

Single rater:

$$ICC(2,1) = \frac{MS_R - MS_E}{MS_R + (k-1)MS_E + \frac{k(MS_C - MS_E)}{n}} = \frac{28.959 - 4.395}{61.927} = \mathbf{0.397}$$

Mean of k raters — **this is the reliability of the gold standard**:

$$ICC(2,k) = \frac{MS_R - MS_E}{MS_R + \frac{MS_C - MS_E}{n}} = \frac{24.564}{30.791} = \mathbf{0.798}$$

Consistency form, which removes per-rater severity offsets: $ICC(3,k) = (MS_R - MS_E)/MS_R = 0.848$. The gap 0.050 is disagreement about *scale use*, not about which pairs match better.

Precision of one MOS. With pooled $SD = 3.163$ and $k = 6$:

$$95\%\text{ CI} = \pm 1.96\,\frac{SD}{\sqrt{k}} = \pm 1.96 \times \frac{3.163}{\sqrt{6}} = \pm 2.53 \text{ points}$$

---

7. Binarising the MOS

$$y_i = \mathbb{1}[\mu_i \geq 5.0]$$

Pair 15: $\mu = 7.571 \geq 5 \Rightarrow y = 1$ (positive).

$$\textstyle\sum_i y_i = 144 \text{ positives}, \qquad 106 \text{ negatives}$$

---

8. Classification and the confusion matrix

$\hat{y}_i(t) = \mathbb{1}[s_i \geq t]$. Pair 15 at $t = 0.66$: $s = 0.6547 < 0.66 \Rightarrow \hat{y} = 0$ while $y = 1$ — a **false negative**.

	$t = 0.66$ (shipped)	$t = 0.6956$ (adopted)
TP — shown, humans agree	138	99
FP — shown, humans disagree	65	25
FN — hidden, humans say yes	6	45
TN — hidden, humans agree	41	81
total	**250**	**250**

---

9. The metrics

$$P = \frac{TP}{TP+FP}, \quad R = \frac{TP}{TP+FN}, \quad F_1 = \frac{2PR}{P+R}, \quad FPR = \frac{FP}{FP+TN}$$

At $t = 0.66$:

$$P = \frac{138}{203} = 0.6798, \quad R = \frac{138}{144} = 0.9583, \quad F_1 = 0.7954, \quad FPR = \frac{65}{106} = 0.6132$$

At $t = 0.6956$:

$$P = \frac{99}{124} = 0.7984, \quad R = \frac{99}{144} = 0.6875, \quad F_1 = 0.7388, \quad FPR = \frac{25}{106} = 0.2358$$

---

10. Choosing the threshold — Youden's J

The pre-registered selection rule is $\arg\max_t J(t)$, with

$$J(t) = R(t) - FPR(t)$$

J was chosen over F_1 because F_1 is inflated by the positive base rate (57.6% here), whereas J is not.

$$J(0.66) = 0.9583 - 0.6132 = 0.3451 \qquad J(0.6956) = 0.6875 - 0.2358 = \mathbf{0.4517}$$

Evaluated at all **246** distinct observed cosine values. The maximum is at

$$t^* = 0.695598304271698$$

which is reported as 0.6956. *Note:* comparing against the rounded literal `0.6956` excludes the very pair that defines the threshold and yields $TP = 98$ rather than 99; the exact float must be used.

---

11. Area under the ROC curve

Via the Mann-Whitney U statistic:

$$AUC = \frac{U}{n_+ \, n_-} = \frac{12466}{144 \times 106} = \mathbf{0.8167}$$

Interpretation: given one aligned and one unaligned pair drawn at random, the aligned pair has the higher cosine 81.7% of the time.

---

12. Uncertainty — bootstrap

Resample the 250 pairs with replacement; recompute Steps 5–10; repeat $B = 3000$ times; take percentiles of the resulting t^*.

$$\text{median } t^* = 0.6956, \qquad 95\%\text{ CI} = [\,0.6666,\ 0.7363\,]$$

The interval **excludes the deployed value 0.66**: only 0.03% of resamples selected a threshold that low.

---

13. Sensitivity — the strict claim

Repeat Steps 7–10 with $y_i = \mathbb{1}[\mu_i \geq 7.0]$ (60 positives), corresponding to the assertion *"this course **teaches** this skill."*

t	Precision	Recall	Cells
0.66	0.286	0.967	203
0.7557	0.667	0.633	57

The strict claim is unsupported at every threshold tested. Displayed cells must therefore be described as **addressing** a skill, not **teaching** it.

---

14. Result

Table 2 — Threshold calibration against the human reference standard. $n = 250$ pairs, $K = 7$ blind annotators, gold = MOS ≥ 5, $ICC(2,k) = 0.798$, $AUC = 0.817$, bootstrap 95% CI $[0.667, 0.736]$.

	0.66 (shipped)	**0.6956** (adopted)
Precision	0.680	**0.798**
Recall	0.958	0.688
F_1	0.795	0.739
Cells displayed	203	124

Conclusion. The similarity measure orders pairs as humans do ($AUC = 0.817$; mean MOS rises monotonically from 2.27 in the lowest cosine decile to 8.85 in the highest), so the method is validated as a ranking instrument. The deployed threshold, however, sits below the range the data supports: raising it from 0.66 to 0.6956 increases precision from 0.680 to 0.798 at the cost of displaying 124 cells rather than 203.

---

15. Reproducibility notes

1. Cosines are stored at full float precision; 0.66 and 0.6956 are display roundings. All comparisons must use the stored value.
2. Precision is a **census** statistic over displayed cells; recall is an **estimate** from 30 sub-floor controls.
3. 6 raters completed all 250 pairs against a pre-registered adequacy floor of 10. Adding a single rater moved t^* from 0.6579 to 0.6956 and precision at 0.66 from 0.833 to 0.680. **All values are provisional.**
4. Rater scores are averaged raw. Per-rater z-normalisation was not pre-registered; the $ICC(3,k) - ICC(2,k)$ gap of 0.050 indicates its likely effect.